



ARTIFICIAL INTELLIGENCE

Class XI

STUDENT HANDBOOK



Subject Code:843

UNIT 7: Leveraging Linguistics and Computer Science

Title: Leveraging linguistics and computer science	Approach: Interactive/ Collaborative / Activity
Summary: Combining the methods and ideas from computer science with linguistics allows for the improvement of many applications pertaining to language understanding and processing. Significant progress is made in natural language processing, machine translation, sentiment analysis, information retrieval, speech recognition, syntax parsing, computational semantics, and dialogue systems by fusing linguistic theories with computer approaches. Computers can now comprehend, analyze, and produce human language more effectively because to this multidisciplinary approach, which also improves their performance in tasks like sentiment analysis, speech recognition, search engine optimization, translation, and conversation interaction.	
Learning Objectives: 1. Understand the challenges of natural language processing (NLP) and its importance in modern technology. 2. Explore the components and processes involved in NLP, including lexical analysis, syntactical analysis, semantic analysis, discourse integration, and pragmatic analysis. 3. Learn about the applications of NLP in various fields such as sentiment analysis, smart assistants, email filtering, predictive text, document analysis, and automatic summarization.	
Key Concepts: 1. Understanding Human Language Complexity 2. Introduction to Natural Language Processing (NLP) - Emotion Detection and Sentiment Analysis, Classification Problems, Chatbot 3. Phases of NLP 4. Applications of NLP	
Learning Outcomes: 1. Develop a better understanding of the complexities of language and the challenges involved in NLP tasks. 2. Learn new techniques and algorithms for NLP tasks.	
Pre-requisites: Knowledge in language, NLP, Data, AI, ML, NLP	

1. Understanding Human Language Complexity

Linguistics is a field of study that focuses on the strategic application of linguistic principles and practices to meet specific objectives or goals. The field of linguistics is often used in marketing, advertising, communications, education, natural language processing, etc. Understanding how language works (including its structure, semantics and pragmatics, as well as sociolinguistics) allows individuals and organizations to tailor their messages, improve communication and influence behavior.

Understanding human language is difficult, even for people who have grown up with it. Human language is incredibly complex, full of strange expressions that seem to contradict each other, metaphors that require cultural knowledge to understand, and grammatical structures that sometimes turn simple ideas into **tongue-twisters**. Machines require systems that research scientists call **Natural Language Processing**, or NLP, to understand human language.

NLP is a branch of Artificial Intelligence (AI) that allows computers to understand, create, and manipulate human speech. NLP has the capability to query the data with natural-language text or voice. It is also known as “language in”. Most consumers have interacted with NLP. For example, NLP is at the core of the technology behind virtual assistants like ODA, Siri, Cortana, Alexa, etc. NLP can be applied to both written text and speech. Some examples of tools that are powered by NLP are: Web search Email spam filtering Auto-translate text or speech Document summarization Sentiment analysis Grammar/spell checking.

Activity 1: Find five interesting facts about “IBM Project Debater” and write in the space provided below.

2. Introduction to Natural Language Processing

Computers are best at working with structured data, in which everything is neatly grouped and labelled. Unfortunately for machines, human language is anything but structured. You have been using language for most of your life. Your brain accomplishes this through some of the most complicated neural circuitry on Earth. But it is very difficult to create machines that can work with human language.

In NLP, machines segment sentences and extract meaning from “tokens” of human language. Human language is unstructured. Although it is loosely held together by rules of grammar, our language expresses information in many confusing ways. Unlike structured information, which can be arranged in tables or matrices with neatly labeled rows and columns, unstructured information is messy and difficult to understand. To see why, consider this famous joke by Groucho Marx.

“One morning I shot an elephant in my pajamas. How he got in my pajamas, I don’t know.”

-Adapted from Groucho Marx, 20th century comedian and movie star

To deal with the “messiness” of unstructured information, computers begin with one sentence at a time. This is called **sentence segmentation**. Computers then break the information into small chunks of information, called **tokens**, that can be individually classified. Once the tokens in text have been sorted into a structure based on what they mean, NLP can work with them.

The following activities show you how Groucho Marx’s joke can be tokenized into useful categories called **entities** and **relationships**.

- An **entity** is a noun representing a person, place, or thing. It is not an adjective, verb, or other article of speech.

Activity 2:

Now keeping this in mind, identify the entities in the sentence “**I shot an elephant in my pajamas.**”

- A **relationship** is a group of two or more entities that have a strong connection to one another.

Activity 3:

Recall Groucho Marx’s quote:

“One morning, I shot an elephant in my pajamas. How he got my pajamas, I don’t know.”

Now keeping this in mind, identify the relationship between the entities in Groucho Marx’s statement:

I + elephant, I + pajamas, in + pajamas, elephant + pajamas, I + shot

Relationship between two entities

Not a relationship between two entities

Once an AI has classified entities and relationships in text or speech, the AI can begin structuring the information as a step toward understanding it. Your brain, by the way, does the same thing, which might have helped you find entities and relationships in the previous activities. For example, consider the following two sentences: “Manu broke the toy. He always breaks toys.” Notice that there is a relationship between the two sentences: the word **he** is related to the word **Manu**. The machine uses NLP to identify this relationship.

- A **concept** is something implied in a sentence but not actually stated. This is trickier because it involves matching ideas rather than the specific words present in the sentence.

Activity 4

Recall, once more, Groucho Marx's statement:

"One morning, I shot an elephant in my pajamas. How he got my pajamas, I don't know."

Based on this statement, select the words that are concepts implied, but not stated.

- ☐ Safari
- ☐ Rifle
- ☐ Photographed
- ☐ Pajamas

Emotion Detection and Sentiment Analysis



Aspect	Emotion Detection	Sentiment Analysis
Definition	Identifies distinct human emotion types.	Measures the strength of an emotion.
Examples	Determining if an expression is anger, happiness, etc.	Assessing if data is positive, negative, or neutral.
Use Cases	Analyzing user ratings, comments in surveys, etc.	Reading social media posts, customer service chats, etc.
AI Training	Can be trained to classify emotions.	Utilizes a sliding scale between positive and negative.
Purpose	Identifying emotional tokens to understand context.	Assessing the overall tone or sentiment of text data.

Classification Problem

Here is an old-fashioned riddle:

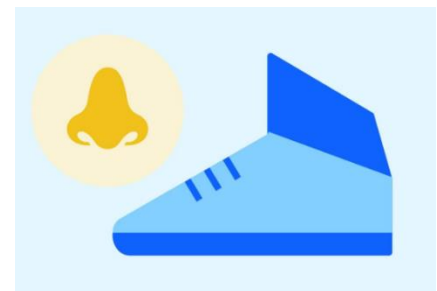
Why does your nose run and your feet smell?

Human language is full of terms that are vague or have double meanings. This is called a **classification problem**.

In this riddle, the phrases "your nose run" and "your feet smell" are used in a humorous way to highlight the ambiguity of language, which poses a classification problem. For instance, "a runny nose" typically refers to having a cold and needing tissues to wipe one's nose, while "a smelly foot" indicates that one's foot has an unpleasant odour. The challenge arises because the words "run" and "smell" have multiple meanings.

Similarly, in everyday language, phrases like "**shipping a box by train**" or "**filling in a form by filling it out**" may seem contradictory or confusing due to the double meanings of the words used. While humans can quickly grasp the intended meaning based on context, AI systems may struggle to accurately classify such phrases without a comprehensive understanding of language nuances and context.

To address this problem,



- An AI system utilizes machine learning techniques such as supervised learning.
- By feeding the system with a large dataset containing examples of language usage and their corresponding classifications, the AI system learns patterns and relationships between words, phrases, and their meanings.
- Over time and with exposure to more data, the AI system improves its classification accuracy by adjusting its internal parameters based on the observed patterns.
- AI systems may not achieve perfect classification accuracy. There will always be some degree of uncertainty or error associated with the system's classifications. To address this, well-designed AI systems not only provide a response but also a confidence value, indicating the system's level of certainty in its classification.

Chatbots

Chatbots are software applications or computer programs designed to simulate conversation with human users, typically through text-based or voice-based interactions. They use artificial intelligence (AI), natural language processing (NLP), and machine learning techniques to understand user queries and provide appropriate responses. Chatbots can be integrated into various platforms such as websites, messaging apps, and voice assistants, and they serve a wide range of purposes including customer support, information retrieval, task automation, and entertainment.

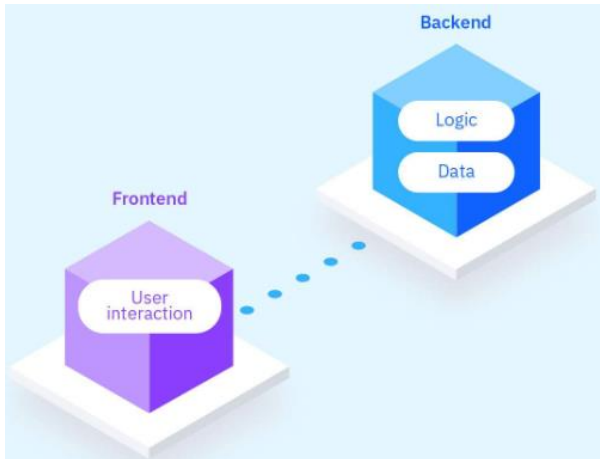
Chatbots, utilized in various fields from retail to healthcare, offer constant availability for online engagement. While not always capable of providing answers, they serve to listen and respond to repetitive inquiries, relieving businesses from the need to allocate human resources for such tasks. *Chatbots operate effectively with small-scale data*, catering to specific needs like movie queries for a cinema chain or broader inquiries for AI systems scanning social media.

Chatbots	Rule-based Chatbots	AI-powered Chatbots
Description	Operate on predefined rules and decision trees. Follow programmed rules to respond to user input.	Utilize natural language processing (NLP) and machine learning algorithms. Also known as chat agents or virtual assistants.
Advantages	- Easy to develop and maintain. - Provide consistent and accurate answers to specific questions.	- 24/7 availability for immediate and consistent support. - Offer personalized interactions based on user preferences and history. - Improve efficiency and cost savings by automating tasks and reducing service costs.
Limitations	- Struggle with understanding complex language. - Unable to adapt to situations beyond programmed rules.	- High development costs and resource requirements. - Prone to biases from training data and lack of transparency in decision-making. - Ethical considerations regarding privacy, manipulation, and responsible use.

Chatbots	Rule-based Chatbots	AI-powered Chatbots
Use Cases	- Customer service tasks like answering common questions and providing order updates. - Guiding users through specific processes.	- Entertainment and Gaming: Engage users with interactive stories and personalized gaming experiences. - Finance and Banking: Answer queries about accounts, transactions, and financial products, and process simple requests.

Structure of a chatbot

A chatbot has a “frontend” and a “backend”.



- The **frontend** of a chatbot serves as the messaging channel through which users interact, providing a user-friendly interface. However, one limitation of the frontend is that it may lack contextual understanding, meaning it might struggle to grasp the full meaning or context of user messages beyond the immediate input.
- The **backend** of a chatbot is where the hard work takes place. The backend operates

application logic and has enough memory to remember earlier parts of a conversation as dialog continues.

Chatbots **understand** a question by breaking it into parts and relating those parts to things in its memory. A chatbot’s goal is to identify **entities** and **intents**, then use what it is found to trigger a **dialog**. Let us understand what intent means.

Intent

An intent is a purpose: the reason why a user is contacting the chatbot. Think of it as something like a **verb**: a kind of action. Users may have various intents when interacting with a chatbot, such as filing a complaint, asking for directions, or speaking to a salesperson. Institutions often have multiple intents that they want their chatbots to address.

For instance, imagine you are tasked with creating a chatbot for a restaurant chain. One potential intent could be to inquire about the restaurant's operating hours. To ensure the chatbot can effectively handle this intent, you might interview a staff member who frequently fields such queries over the phone. Then, your goal would be to compile a comprehensive list of the diverse ways in which customers might inquire about the restaurant's opening hours. The following table provides many examples of possible user inputs that map to this kind of intent.

Intent	Possible user inputs
Open	When do you open?
	What are your hours?
	You open now?
	How late are you open?
	Can I walk in at 7 pm?

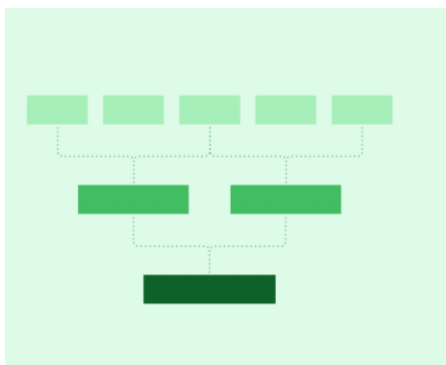
Entity

An entity is a noun: a person, place, or thing. Once you have a list of the intents you want your chatbot to fulfill, you are ready to continue. If a user asks, “What are the hours for the Bangalore office?”, then providing business hours is the **intent** and Bangalore is the **entity**. A chatbot needs a full list of entities to be helpful. The following table lists examples of entities that map to the intent and possible user inputs of the previous restaurant chain example.

Intent	Possible user inputs	Entities
Open	When do you open?	Bangalore
	What are your hours?	Schedule
	You open now?	Time
	How late are you open??	Time
	And so forth; there are many inputs mapping to this intent.	And so forth; there are many entities implied by this intent.

Dialog

A **dialog** is a flowchart—an IF / THEN tree structure that illustrates how a machine will respond to user intents. A dialog is what the machine replies after a human asks a question. Even if a human uses run-on sentences, poor grammar, chat messaging expressions, and so on, artificial intelligence allows the NLP to understand well enough to provide a response.



The dialog represents each possible word or phrase a user might enter, the matched response for the chatbot, and the many possible subsequent replies a user might make next. That is too much for an ordinary flowchart to show (you might need three or four dimensions!), so chatbot software condenses each moment of the conversation into a node. A node contains a statement by the chatbot and a long, expandable list of possible replies.

Creating the flowchart for a chatbot's responses entails a comprehensive process. Each potential user input following the chatbot's initial greeting must be assigned a corresponding reply. For instance, in the case of inquiries about a restaurant's operating hours, all conceivable questions would trigger a single response. This mapping of a vast array of possible questions to a limited number of responses persists throughout the conversation until its conclusion.

Natural Language Processing – Converting Speech to Text & analysing its intent

Natural language processing (NLP) involves a series of five phases that enable machines to analyse, categorize, and understand both spoken and written language. These steps utilize deep neural network-style machine learning techniques to mimic the brain's ability to process data accurately. Businesses leverage tools and algorithms aligned with these NLP stages to extract insights from vast amounts of data and facilitate informed decision-making. Effective comprehension of these NLP stages is essential for optimizing the use of NLP in text and voice applications. Let's explore each step-in detail:

1. Lexical analysis

This step involves understanding and examining the structure of words in a language. It breaks down the text into paragraphs, phrases, and words. Lexical normalization techniques like stemming and lemmatization are commonly used to reduce words to their base forms.

- Stemming reduces words to their root form, such as removing suffixes like "ing", "ly", "es", and "s".
- Lemmatization reduces words to their dictionary form, considering factors like parts of speech (POS) to determine their meaning in context.

2. Syntactical Analysis

Syntactic Analysis is used to check grammar, word layouts, and word relationships.

Example: Mumbai travels to the Anuj.

The line "Mumbai travels to Anuj" makes no sense, hence it is rejected by the Syntactic Analyzer. **Syntactical parsing** is the analysis of words in a sentence for grammar. Dependency Grammar and Part of Speech (POS) tags are significant syntactic elements.

3. Semantic Analysis

Semantic analysis aims to understand the various meanings conveyed by a sentence in a clear and contextually appropriate manner. It extracts relevant insights from the text to comprehend its intended message.

4. Discourse Integration

This involves understanding the context of a statement or word based on preceding sentences or words. It helps interpret references like pronouns and proper nouns by identifying their connections with earlier parts of the conversation.

Example- Arti wants it.

We can observe from the following sentence that the "it" keyword makes no sense. In reality, it applies to anything we don't know. That is all this "it" word depends on the prior sentence, which is not provided. So, if we know what "it" is, we can simply find the reference.

5. Pragmatic Analysis

It denotes the study of meanings in a particular language. Process of extracting insights from a text. It involves verbal repetition, such as "who said what to whom?" It recognizes how individuals communicate with one another, the context in which they are speaking, and many other factors.

Video Session: Five Phases of NLP (Natural Language Processing) (with Tools and Applications) -

<https://www.youtube.com/watch?v=8SSAliFwCy8>

Applications of NLP

Businesses can use natural language processing tools to analyse data, discover insights, automate time-consuming operations, and gain a competitive advantage.

- **Sentiment Analysis:** Natural language processing (NLP) may evaluate consumer comments, social media posts, product reviews, and other text data to determine the sentiment (positive, negative, or neutral) associated with a specific brand, product or service. This data aids firms in understanding client impressions and sentiments.
- **Voice Assistants:** Voice assistants are becoming increasingly popular! Whether it's Siri, Alexa, or Google Assistant, practically everyone uses one to make calls, create reminders, plan meetings, set alarms, browse the internet, and so on. How do they work? They use a complicated blend of voice recognition, natural language interpretation, and natural language processing to understand and respond to what humans say. The long-term purpose of voice assistants is to serve as a bridge between humans and the internet, providing a variety of services through simple speech interaction. However, they are still a long way from achieving that aim, as Siri occasionally fails to grasp what you are saying!
- **Email Filtering:** Email is a part of our daily lives. We find ourselves bombarded with emails about job, study, and a variety of other topics. We receive emails from a variety of sources; some are work-related or from our dream school or institution, while others are spam or promotional in nature. Here, Natural Language Processing comes into play. It classifies incoming emails as "important" or "spam" and assigns them accordingly.
- **Document Analysis:** Document analysis is another use of natural language processing. Companies, institutions, and schools, among other places, are constantly inundated with data that must be properly organized, stored, and searched. All of this may be accomplished with NLP. It not only searches a keyword but also categorizes it according to the instructions, saving us from the tedious and time-consuming task of searching for a single person's information from a large number of files. It is not only restricted to this, but it also assists users in making informed decisions about claims and risk management.
- **Automatic Summarization:** Data has grown in line with technological advancements. This rise of data has broadened the scope of data processing. Still, manual data processing is time-consuming and error-prone. NLP provides a solution for this as well; it can not only summarize the meaning of information but also identify the emotional meaning hiding within it. As a result, the summary process becomes more efficient and precise.

Activity: Check out <https://sites.research.google/versebyverse/>

This is an experimental AI-powered muse that helps you write poetry inspired by classic American poets!

Activity: Creating a Chatbot

Create a chatbot on ordering ice-creams using any of the following platforms:

- Google Dialogflow
- Botsify.com
- Botpress.com

Video session (for Google Dialogflow) : <https://www.youtube.com/watch?v=bIXkqDZMgaI>

Activity: For advanced learners. The following program prints the POS tags of a statement.

```
pip install nltk
```

```
import nltk
from nltk.tokenize import word_tokenize
from nltk import pos_tag

nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')

# Sample sentence
sentence = "The quick brown fox jumps over the lazy dog."

# Tokenize the sentence into words
words = word_tokenize(sentence)

# Perform part-of-speech tagging
pos_tags = pos_tag(words)

# Print the tokens with their respective parts of speech
for word, pos in pos_tags:
    print(f"Word: {word}, POS: {pos}")
```

Output:

```
Word: The, POS: DT
Word: quick, POS: JJ
Word: brown, POS: NN
Word: fox, POS: NN
Word: jumps, POS: VBZ
Word: over, POS: IN
Word: the, POS: DT
Word: lazy, POS: JJ
Word: dog, POS: NN
Word: ., POS: .
```

Activity: For Advanced Learners Creating a simple rule based chatbot using Python

```
import random

def get_response(user_input):
    user_input = user_input.lower()
    if "hello" in user_input:
        return "Hi there! How can I assist you?"
    elif "how are you" in user_input:
        return "I'm just a bot, but thanks for asking!"
    elif "bye" in user_input:
        return "Goodbye! Have a great day!"
    else:
        return "I'm sorry, I didn't understand that."

def main():
    print("Welcome to the Simple Chatbot!")
    print("Type 'bye' to exit.")

    while True:
        user_input = input("You: ")
        if user_input.lower() == 'bye':
            print("Chatbot: Goodbye! Have a great day!")
            break
        else:
            response = get_response(user_input)
            print("Chatbot:", response)

if __name__ == "__main__":
    main()
```

Output:

```
Welcome to the Simple Chatbot!
Type 'bye' to exit.
You: hello good afternoon
Chatbot: Hi there! How can I assist you?
You: how are you
Chatbot: I'm just a bot, but thanks for asking!
You: bye
Chatbot: Goodbye! Have a great day!
```

EXERCISES

A. Multiple Choice Questions:

1. Which of the following is NOT a common task in NLP?
 - a) Machine translation
 - b) Text summarization
 - c) Speech recognition
 - d) Image recognition
2. What is the main challenge/s of NLP?
 - a) Handling Ambiguity of Sentences
 - b) Handling Tokenization
 - c) Handling POS-Tagging
 - d) All of the mentioned
3. What is a chatbot?
 - a) A physical robot used for chatting purposes.
 - b) A computer program designed to simulate conversation with human users, especially over the internet.
 - c) An advanced form of search engine.
 - d) A tool used for sending automated emails.
4. Which of the following is an application of Natural Language Processing (NLP)?
 - a. Autonomous vehicles
 - b. Predicting stock prices
 - c. Sentiment analysis
 - d. Virtual reality gaming
5. Which of the following statements about Voice Recognition Interfaces is true?
 - a. They solely rely on text-based inputs.
 - b. They are incapable of understanding multiple languages.
 - c. They convert spoken language into text or commands.
 - d. They require a physical keyboard for interaction.

B. Short answer questions:

1. How does NLP help in email filtering? Give a real-life example.
2. List the steps of NLP Processing.
3. Briefly explain the two types of chatbots.
4. Briefly explain the classification problem. Give at least two examples.
5. Define the following:
 - (a) Intent
 - (b) Entity
 - (c) Dialog

C. Long Answer Questions:

1. Explain the structure of a chatbot.
2. "A syntax tree is created as part of the procedure to visually represent semantic links." Identify the phase of NLP processing?

D. Case Study questions:

1. Imagine you are a customer service manager at a global e-commerce company facing increasing customer inquiries across multiple channels. To alleviate the strain on your support team and enhance customer satisfaction, you decide to implement a chatbot solution. Develop a case study outlining the challenges you faced, the criteria you used to select a chatbot platform.
2. Imagine you are a customer experience lead at a telecommunications company, currently relying on a rule-based chatbot to handle customer queries. However, due to limitations in scalability and adaptability, you are considering a transition to an AI-based chatbot solution. Detail the challenges faced during the transition, and the observed impact on customer service efficiency and satisfaction.